The model pinned to the recordings, and the comparison

> **QUESTION**
>
> When the block model is pinned to the recordings, how well does it predict which odor pairs become more similar when APL is blocked, and does APL's specific wiring improve that prediction?

INPUTS

- The model of Report 05 (hemibrain R, male CNS L and R; corrected calyx, new claws, new ORN table)
- The recordings and the quantities of Report 06

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/odor_weighted/` : `TNT_vs_model_fits.csv` , `TNT_vs_model_nulls.csv` , `TNT_vs_model_odor_pairs.csv` , `TNT_vs_model_per_odor.csv` , `TNT_vs_model_summary.json` , `TNT_vs_model.png`

SCRIPTS

- `scripts/tnt_compare.py`
- `scripts/reports/r07_model_vs_tnt.py` (this report)

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Report 05 ran the model with assumed coding levels (20% of KCs respond with APL blocked, 10% intact). Here the model is instead **pinned** to the recordings: the fraction of KCs responding to each odor with APL blocked is set to the TNT fraction for that odor, and the mean intact fraction to TNT3c's. The model's change in similarity is then compared with the data's, pair by pair. The pinning sets coding levels only. The pattern of similarity changes across pairs is not fitted, so it is a prediction.

What changed in this revision (2026-11-04)

The model is rebuilt on the corrected calyx, the new claws and the new ORN table; version MC (per-KC counts plus bouton inhibition per glomerulus) is added.

Equation 1 (the input gain g_o) is worked through with a mock example.

New: why the ratio to the ceiling is squared, with a simulation.

The suppression figure has a y = x line and equal axes.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
x_{ko}, y_{ko}, A_o, a, u_k, v_k, p_g	As in Report 05	
θ	KC threshold, fixed at 1 here (the scale is carried by g_o)	1
g_o	Input gain of odor o: multiplies every glomerulus's response to o so that the blocked model matches TNT	one per odor, mushroom body and input table

Symbol	What it stands for	Values, units, range
ϕ_o^{TNT}, ϕ_o^{TNT3c}	Fraction of units responding (> 1) in the data (Report 06)	0 to 1
$Q_o(q)$	The q-quantile of the drive $\sum_g W_{kg} r_{go}$ over KCs	response units
$\Delta S_{ab}^{\mathrm{data}}$	$S_{ab}^{\mathrm{TNT}} - S_{ab}^{\mathrm{TNT3c}}$ (Report 06)	difference in cosine
$\Delta S_{ab}^{\mathrm{model}}$	Model similarity blocked − intact, each averaged over the three mushroom bodies	difference in cosine
r	Pearson correlation over odor pairs between ΔS^{data} and $\Delta S^{\mathrm{model}}$	−1 to 1
s_o	Suppression of odor o: blocked ÷ intact fraction responding (data or model)	×
ρ	Spearman correlation over odors between data and model suppression	−1 to 1
p	Probability under a null (odor labels permuted, or wiring shuffled) of a value at least as large	0 to 1

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1 Step 1: pinning each odor to the TNT fraction

With APL blocked the model is $y_{ko} = [x_{ko} - \theta]_+$. Instead of one threshold for all odors, each odor's glomerular responses are multiplied by a gain g_o, with $\theta = 1$. The gain is chosen so that exactly the TNT fraction of KCs exceeds the threshold:

$$x_{ko} = g_o \sum_g W_{kg}\, r_{go}, \qquad g_o = \frac{\theta}{Q_o(1 - \phi_o^{\mathrm{TNT}})} \tag{1}$$

$Q_o(q)$ = the q-quantile over KCs of $\sum_g W_{kg} r_{go}$; dividing by it puts the KC at that quantile exactly at threshold

So an odor that drives 25% of units in TNT drives 25% of model KCs with APL blocked. Only the order of the KCs' drives comes from the connectome and the ORN (or GH146) responses; the level comes from the data. Then, with APL intact, a single APL gain a per mushroom body and version is calibrated so that the mean fraction over odors equals TNT3c's (6.83% for the 16 odors). How much each odor is suppressed is left to the model.

1.1 Equation 1 with numbers: a mock example

The goal of pinning is that, with APL blocked, the model drives exactly the fraction of KCs that TNT drives for each odor. The threshold is fixed at θ = 1, and each odor's input is multiplied by a gain g_o chosen so that the right number of KCs end up above 1. Here is the calculation for a mock mushroom body of 10 KCs and one odor that drives 20% of units in TNT:

- **1. Unscaled drives** $\sum_g W_{kg} r_{go}$ of the 10 KCs: 0.12, 0.30, 0.05, 0.42, 0.22, 0.18, 0.60, 0.35, 0.08, 0.50.
- **2. Sort them:** 0.05, 0.08, 0.12, 0.18, 0.22, 0.30, 0.35, 0.42, 0.50, 0.60.
- **3. The quantile.** 20% should respond, so the threshold must sit at the 80th percentile, $Q_o(1 - 0.20) = Q_o(0.80)$. With 10 values its position is 0.80 × 9 = 7.2 places from the bottom (counting from 0): between the 8th value (0.42) and the 9th (0.50), 20% of the way: 0.42 + 0.2 × (0.50 − 0.42) = **0.436**.
- **4. The gain.** $g_o = \theta / Q_o(0.80) = 1/0.436$ = **2.294**.

- **5. Scaled drives** $x_{ko} = g_o \sum_g W_{kg} r_{go}$: 0.28, 0.69, 0.11, 0.96, 0.50, 0.41, 1.38, 0.80, 0.18, 1.15. Exactly two (from 0.60 and 0.50) are above 1, so 2 of 10 = 20% respond, as in TNT. The KC at 0.42 ends just below, at 0.96.

Dividing by the quantile only moves the threshold to the right place in each odor's distribution. The order of the KCs' drives (which KC responds) still comes from the connectome and the ORN responses. An odor with weak ORN responses has small drives, a small quantile, and therefore a large gain (Figure 1b).

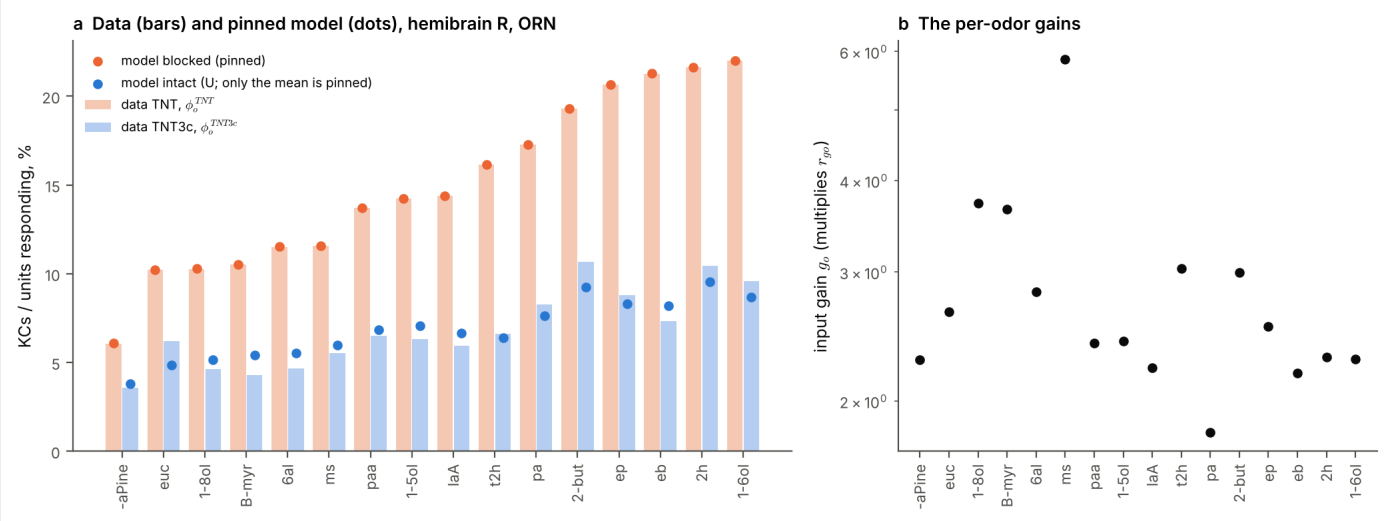


Figure 1. Pinning the model to the recordings. a, For each odor (sorted by the TNT fraction), the percentage of units responding in the data (bars: orange TNT, blue TNT3c) and of model KCs responding (dots), hemibrain R with ORN input and uniform APL. Orange dots sit on the bars by construction. Blue dots are free per odor; only their mean is set to the TNT3c mean. **b**, The gain g_o that pins each odor (log scale). Odors with weak ORN responses but many TNT responders need large gains.

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2 Step 2: the APL versions compared

All versions of Report 05 are run with the same pinning, plus two in which APL's activity is fed forward from the PNs instead of fed back from the KCs:

$$\text{F0: } y_{ko} = \left[x_{ko} - 1 - a\, A_o^{\text{ff}}\right]_+, \quad A_o^{\text{ff}} = \frac{\sum_g w_g\, g_o r_{go}}{\sum_g w_g}; \qquad \text{F: the same with } w_g \text{ replaced by } w_g\, \tilde{p}_g^{\text{in}} \tag{2}$$

$w_g = \sum_k W_{kg}$ (total claw input of glomerulus g); $\tilde{p}_g^{\text{in}}$ = PN→APL per PN→KC synapse, scaled to a claw-input-weighted mean of 1 (Report 04)

In F0, APL's activity is the odor's mean KC input, so it does not depend on how many KCs respond. In F, glomeruli with more PN→APL synapses contribute more to it.

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3 Step 3: how model and data are compared

- **Odor-pair pattern**, r: the Pearson correlation over odor pairs between $\Delta S_{ab}^{\text{data}}$ and $\Delta S_{ab}^{\text{model}}$. The model value is averaged over the three mushroom bodies before the correlation. Every pair counts (120 with ORN input, 78 with GH146).
- **Odor-label permutation** p: the odor labels of the model matrix are permuted 5,000 times. This keeps the matrix structure and tests whether the match depends on which odor is which.
- **Bootstrap range**: recordings are resampled with replacement within each condition (2,000 draws), and the 5th–95th percentile of r is reported.

- **Wiring nulls**: each wiring version is re-solved with its wiring shuffled 200 times per mushroom body. C swaps (u_k, v_k) pairs among KCs of the same class. P and M shuffle p_g across glomeruli. F shuffles $\tilde{p}_g^{\text{in}}$. p is the share of shuffles with r at least as high as the real wiring.
- **Ceiling**: because the data are noisy, even a perfect model would not reach $r = 1$. The split-half reliability of ΔS^{data} (Spearman–Brown corrected) is 0.70, which caps r near $\sqrt{0.70}$ = 0.84 (explained in Section 4).
- **Suppression**: Spearman ρ over odors between data and model s_o.

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4 Results

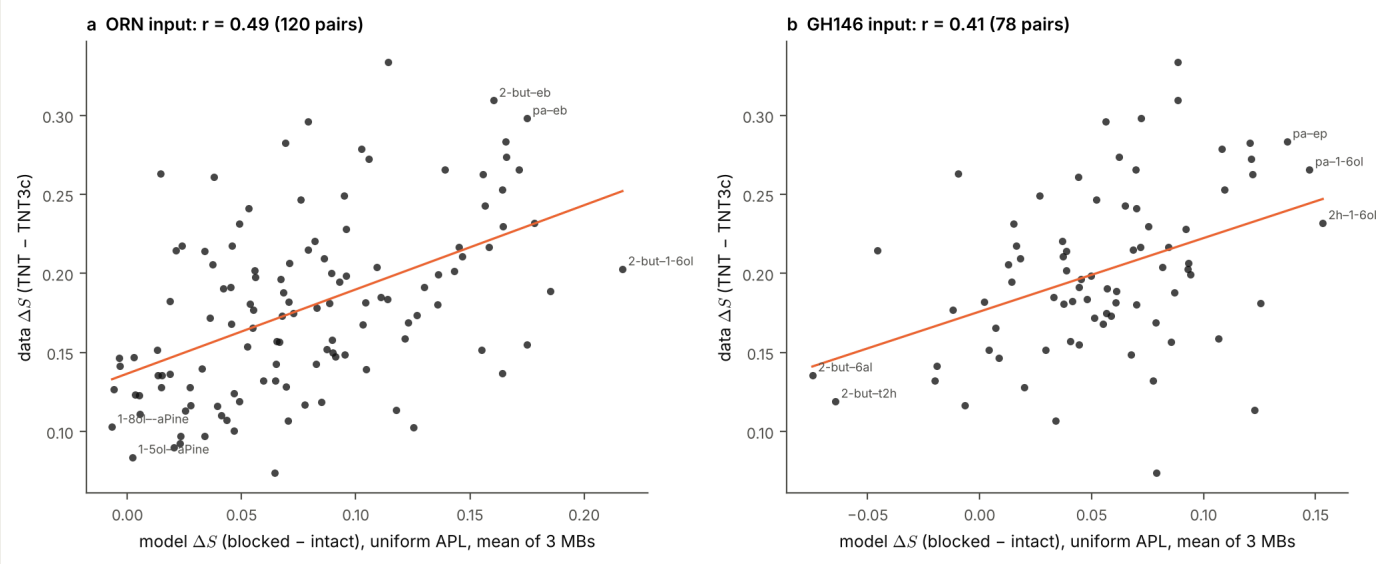


Figure 2. Model against data, pair by pair. Each point is one odor pair. Horizontal: the model's change in similarity when APL is blocked (uniform APL, pinned, mean of hemibrain R and male CNS L and R). Vertical: the data's change, TNT − TNT3c. The orange line is the least-squares fit. Note the different axis ranges: the model's change is smaller than the data's.

With uniform APL, r = 0.49 with ORN input (permutation p = 0.001; bootstrap 0.28–0.53) and 0.41 with GH146 input. Against the ceiling of 0.84, that is about $(0.49/0.84)^2 \approx 35\%$ of the reproducible pattern (why squared: below). Per mushroom body r is 0.39 (hemibrain R), 0.56 (male CNS L), 0.45 (male CNS R).

4.1 Why the ratio to the ceiling is squared

A correlation r is not a share. Its square, r^2, is: the share of the data's variance that the model accounts for. The reasoning goes in three steps.

- **The data are part pattern, part noise.** The split-half reliability of ΔS is 0.702: 70% of its variance over pairs is reproducible pattern, and 30% is recording noise that changes from one set of recordings to the next.
- **A perfect model can explain only the pattern.** Its r^2 would be 0.702, so its r would be $\sqrt{0.702}$ = 0.838. That is the ceiling.
- **The uniform model.** r = 0.49, so r^2 = 0.244: it accounts for 24% of all the variance, which is 35% of the 70% that is pattern. That is $r^2/\text{reliability} = (r/\text{ceiling})^2$.

A quick simulation confirms it: 120 mock pairs whose "data" are 70% pattern and 30% noise, and a mock model that shares 35% of its variance with the pattern. The model correlates with the noisy data at r = 0.49 on average and with the noise-free pattern at r = 0.58 (0.59^2 = 0.35). The unsquared ratio 0.49/0.84 = 0.59 is that second correlation; squaring turns it into a share.

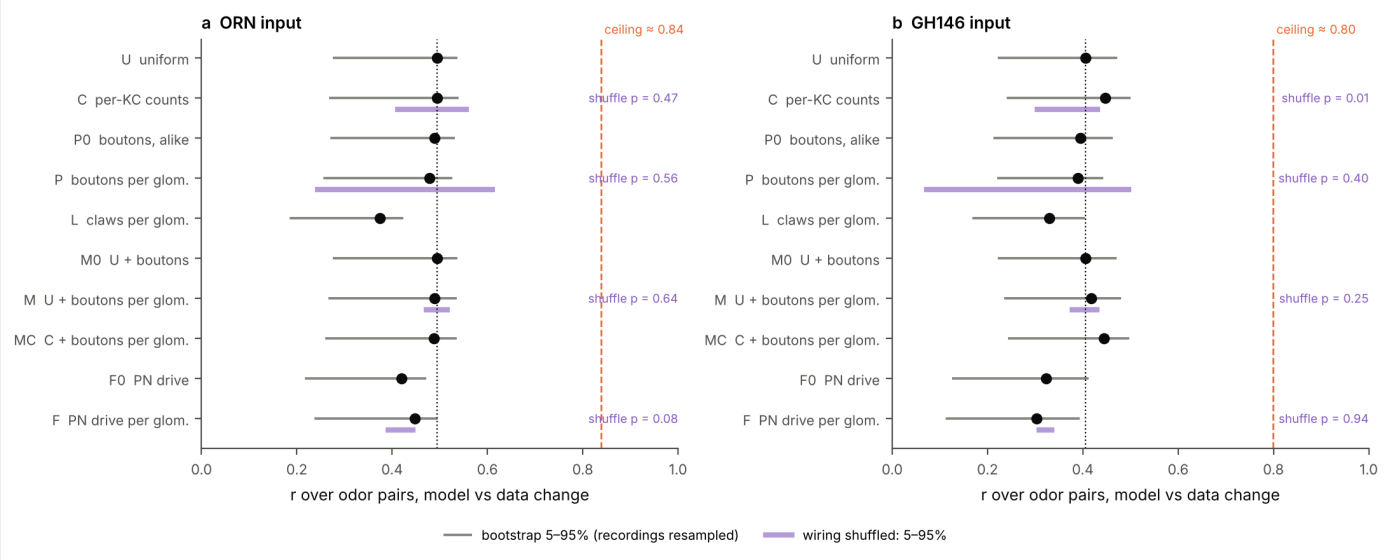


Figure 3. Every version against the data. For each APL version, black dots are r between the model's and the data's change in similarity over odor pairs. Grey lines are the bootstrap 5–95% range over recordings. Purple bars are the 5–95% range of r when that version's wiring is shuffled (200 shuffles per mushroom body), and p is the share of shuffles that do at least as well as the real wiring. The dotted line marks uniform APL; the dashed orange line marks the ceiling set by the data's noise.

Table 2. Fit of every version (threshold 1). Data mean ΔS = 0.178 (16 odors). Shuffle p only for the 4 wiring versions with a null (C, P, M, F).

version	r ORN	vs U	shuffle p	r GH146	vs U	shuffle p	suppression ρ ORN	mean model ΔS (ORN)
U uniform	0.49	+0.00		0.41	+0.00		0.29	0.078
C per-KC counts	0.49	−0.00	0.47	0.45	+0.04	0.01	0.32	0.060
P0 boutons, alike	0.49	−0.01		0.39	−0.01		0.31	0.079
P boutons per glom.	0.48	−0.02	0.56	0.39	−0.02	0.40	0.29	0.082
L claws per glom.	0.37	−0.12		0.33	−0.08		0.35	0.073
M0 U + boutons	0.49	−0.00		0.41	−0.00		0.32	0.078
M U + boutons per glom.	0.49	−0.00	0.64	0.42	+0.01	0.25	0.34	0.079
MC C + boutons per glom.	0.49	−0.01		0.44	+0.04		0.35	0.063
F0 PN drive	0.42	−0.07		0.32	−0.08		0.19	0.084
F PN drive per glom.	0.45	−0.05	0.08	0.30	−0.10	0.94	0.31	0.085

- **Feedback matters.** Uniform feedback gives r = 0.49; inhibition fed forward from the PNs (F0, F) gives r = 0.42 and 0.45. Report 08 explains why.
- **Specific wiring.** The wiring versions (C, P0, P, M0, M, MC) change r from U's by −0.02 to −0.00 with ORN input, and −0.02 to +0.04 with GH146. Against their own shuffles, C with GH146 input (p = 0.015,

r = 0.45) passes at $p < 0.05$; every other test does not. L (inhibition on claws per glomerulus) gives r = 0.37 (ORN) and 0.33 (GH146).

- **Suppression per odor is not predicted** (ρ ≤ 0.35).
- **Size.** The model's mean change (+0.078) is 0.44 times the data's (+0.178).

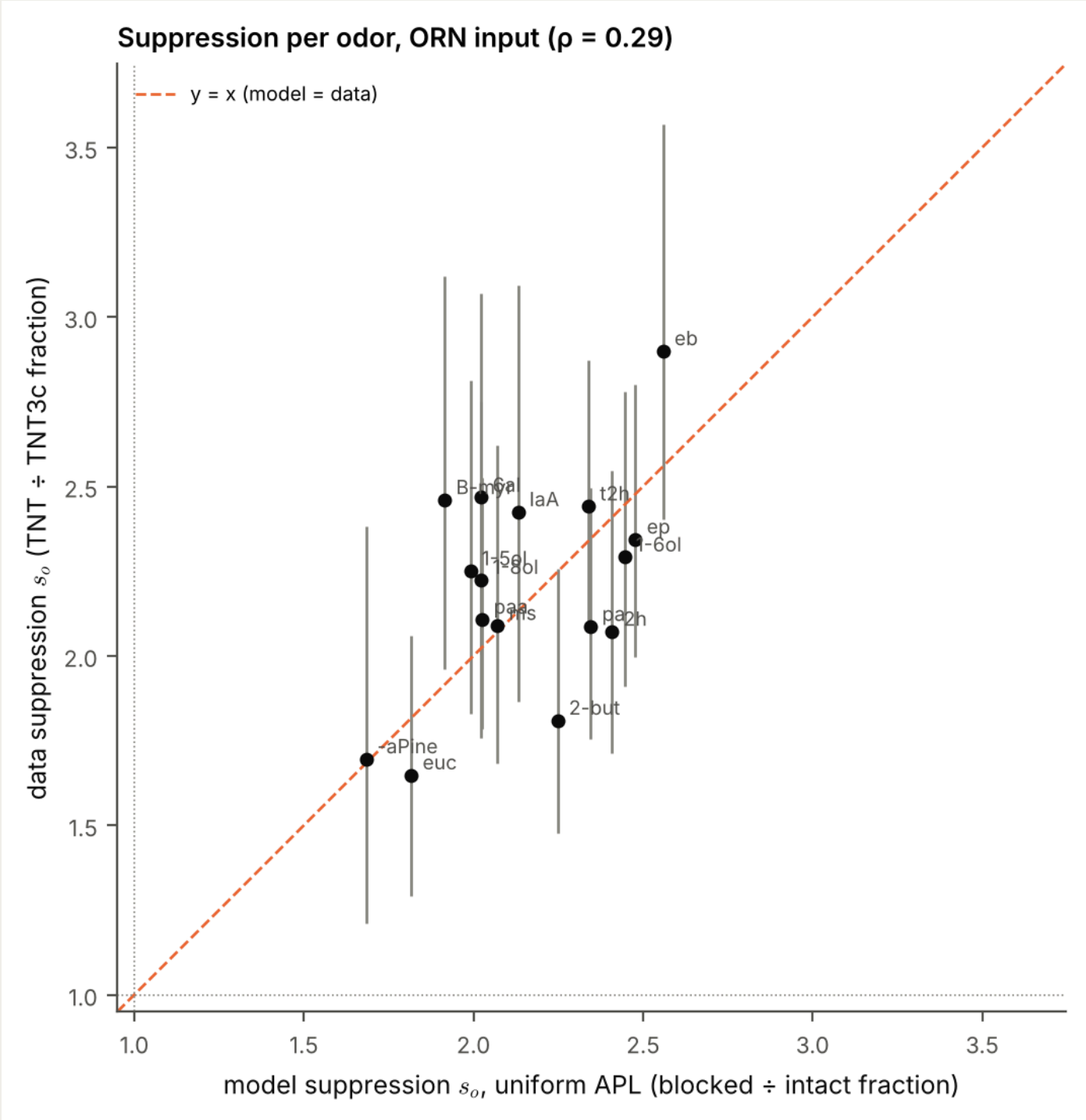


Figure 4. Which odors APL suppresses most. Each point is one odor. Horizontal: how many times more model KCs respond with APL blocked than intact (uniform APL, mean of three mushroom bodies). Vertical: the same ratio in the data, with its bootstrap 5–95% range over recordings (grey). The dashed orange line is y = x, where model and data would agree; the dotted lines mark 1 (no suppression). Both axes use the same scale. 10 of 16 odors lie above the line: the data suppress them more than the model does.

Caveats

Inputs are ORN or GH146 averages from other flies, not the recorded flies' PNs. Imaged units are not identified KCs.

Three mushroom bodies from two connectomes. Their uniform-model fits range from 0.39 to 0.56 (ORN input), so wiring tests that average over them have little power.

Many tests: of 8 real-vs-shuffle tests on odor pairs, one passes at $p < 0.05$: C with GH146 input ($p = 0.015$, r = 0.45). With 8 tests, about 0.4 would pass by chance alone.

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